

NexaCorp Enterprise Knowledge Copilot — User Guide

Welcome to the NexaCorp Knowledge Copilot, a local AI assistant designed to help you quickly query company runbooks, search support tickets, and manage IT issues.

1. Quick Start (How to Launch the App)

If the system has already been set up by your IT administrator, you can open the applications instantly:

Option A: The Visual Web Dashboard (Recommended)

1. Open your web browser (Chrome, Firefox, or Edge).
2. Go to the address: `http://localhost:8501`
3. The visual dashboard will load, showing the chat window and setup settings.

Option B: The API Sandbox (For Programmatic/Fast Testing)

1. Open your web browser.
2. Go to the address: `http://localhost:8000`
3. You will see a fast, interactive portal where you can execute queries and inspect JSON responses instantly.

2. Navigating the Web Dashboard

The main screen is divided into two primary zones: **The Settings Sidebar** on the left, and **The Chat Workspace** on the right.

SETTINGS SIDEBAR	CHAT WORKSPACE
1. USER ROLE SELECTION	User: "How do I fix the VPN error?"
[Employee Manager Admin]	
error..."	Copilot: "Here is how to resolve the VPN
2. SEARCH ENGINE SELECTION	
[Edge Sandbox Enterprise]	[📄 View Sources] (Click to expand)
3. VISUAL CONTROLS	[Confidence: 89%] [Faithfulness: 1.0]
[Relevance: 0.9]	
[x] Show Reasoning Trace	

1. Settings Sidebar (Left Panel)

- **User Role Selector:** Sets your access permissions.
- **Search Engine Selector:** Toggles how the system retrieves documents.
- **Show Reasoning Trace Checkbox:** Reveals the step-by-step decision-making process of the AI.

2. Chat Workspace (Right Panel)

- **Chat Input:** Type your questions here (e.g., *"How do I reset my MFA password?"*).
- **AI Answer:** The response generated by the local AI model.
- **Sources Expander:** Click this to view the exact text chunks of handbooks or tickets used to answer your question.
- **Real-time Metrics Badges:** Transparency scores checking retrieval confidence, faithfulness (absence of hallucinations), and relevance.

3. Core Features & How to Use Them

Feature 3.1 — Smart Role-Based Security

The application filters information based on your job role. You can test this by toggling roles in the sidebar:

- **Employee:** Standard access. Allows reading public handbook rules, runbooks, and generic policies.
- **Manager:** Middle tier. Allows reading organizational layouts, system owner charts, and tracking escalation tickets.
- **IT Admin:** Full access. Allows reading restricted network logs, active system metrics, and security incident reports.

Feature 3.2 — Database Search Engines

You can choose how the Copilot retrieves documents:

- **Edge Sandbox (Local):** A lightweight local index on your computer. Great for offline or small-scale work.
- **Enterprise Cluster (Elasticsearch):** A professional, high-performance database cluster. Scales to handle millions of documents with sub-second response times.

Feature 3.3 — Automatic PII Redaction (Privacy Shield)

To protect employee privacy, the system automatically runs a redaction engine before showing you any search results:

- All real names are replaced with [REDACTED_PERSON] .
- Email addresses are replaced with [REDACTED_EMAIL] .
- Phone numbers are replaced with [REDACTED_PHONE] .
- *You can verify this by looking inside the expanded "Sources" section after asking a question.*

Feature 3.4 — Smart Ticket Creation (Deduplication Guard)

If you ask the Copilot to file a new IT ticket (e.g., *"Create a ticket for the VPN auth error"*):

1. The Copilot automatically scans existing open tickets.

2. If it finds a matching open issue (over **75% similarity**), the system will **block** the creation of a duplicate ticket.
3. It will display a warning:  Potential duplicate detected! and direct you to the existing ticket ID, saving IT support staff from duplicate queues.

Feature 3.5 — Show Reasoning Trace

If you check **Show Reasoning Trace**, the app shows you an expander detailing how the AI classified your query (e.g., Routing to docs search vs. tickets search vs. create_ticket tool).

4. Visual Verification Metrics

Every answer comes with three quality control indicators:

1. **Retrieval Confidence**: Represents the closeness of matches.
 2. **Faithfulness Score (0.0 to 1.0)**: Evaluates if the AI's response is strictly grounded in the retrieved sources. A score of 1.0 means 100% of the facts came directly from the handbook or tickets, guaranteeing **no hallucinations**.
 3. **Context Relevance**: Checks how well the retrieved documents match the intent of your question.
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Appendix: Administrator Installation & Ingestion

(For developers and IT administrators setting up the system from scratch)

A.1 Prerequisites

1. Install **Python 3.11** and **Podman/Docker**.
2. Install **Ollama** and download the LLaMA model:

```
ollama run llama3.2:latest
```

3. Install project requirements:

```
pip install -r requirements.txt
python -m spacy download en_core_web_sm
```

A.2 Starting Local Services

1. **Elasticsearch Container:**

```
podman run -d --name elasticsearch -p 9200:9200 -p 9300:9300 -e
"discovery.type=single-node" -e "xpack.security.enabled=false"
docker.elastic.co/elasticsearch/elasticsearch:8.11.1
```

A.3 Ingesting the 200,000 Tickets

To populate the databases with your corporate knowledge datasets:

- **Ingest to local sandbox:**

```
PYTHONPATH=. python src/ingest_full.py
```

- **Ingest to local Elasticsearch container:**

```
python scripts/ingest_elasticsearch.py
```